

INCREASED CROSS-EDUCATION OF MUSCLE STRENGTH AND REDUCED CORTICOSPINAL INHIBITION FOLLOWING ECCENTRIC STRENGTH TRAINING

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Abstract—Aim: Strength training of one limb results in a substantial increase in the strength of the untrained limb, however, it remains unknown what the corticospinal responses are following either eccentric or concentric strength training and how this relates to the cross-education of strength. The aim of this study was to determine if eccentric or concentric unilateral strength training differentially modulates corticospinal excitability, inhibition and the cross-transfer of strength. **Methods:** Changes in contralateral (left limb) concentric strength, eccentric strength, motor-evoked potentials, short-interval intracortical inhibition and silent period durations were analyzed in groups of young adults who exercised the right wrist flexors with either eccentric ($N = 9$) or concentric ($N = 9$) contractions for 12 sessions over 4 weeks. Control subjects ($N = 9$) did not train. **Results:** Following training, both groups exhibited a significant strength gain in the trained limb (concentric group increased concentric strength by 64% and eccentric group increased eccentric strength by 62%) and the extent of the cross-transfer of strength was 28% and 47% for the concentric and eccentric group, respectively, which was different between groups ($P = 0.031$). Transcranial magnetic stimulation revealed that eccentric training reduced intracortical inhibition (37%), silent period duration (15–27%) and increased corticospinal excitability (51%) compared to concentric training for the

untrained limb ($P = 0.033$). There was no change in the control group. **Conclusion:** The results show that eccentric training uniquely modulates corticospinal excitability and inhibition of the untrained limb to a greater extent than concentric training. These findings suggest that unilateral eccentric contractions provide a greater stimulus in cross-education paradigms and should be an integral part of the rehabilitative process following unilateral injury to maximize the response. © 2015 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: corticospinal inhibition, cross-activation, cross-transfer, ipsilateral motor cortex, strength, recovery.

INTRODUCTION

The potential to increase muscle strength following resistance training is well-documented and overloading skeletal muscle with eccentric strength training has been shown to be superior compared to concentric strength training for increasing muscle strength (Hortobágyi et al., 1996). An interesting observation, originally described by Scripture et al. (1894), is the phenomena of cross-education, whereby strength training of a single limb was found to increase the strength of the untrained limb. Since this initial report, several studies have provided evidence to support the existence of cross-education using concentric, eccentric and isometric strength training (Cannon and Cafarelli, 1987; Brown et al., 1990; Munn et al., 2005; Lee et al., 2009). Interestingly eccentric training produces the largest changes in strength compared to concentric and isometric (Enoka, 1996; Hortobágyi et al., 1997, 1999). However, the mechanism that modulates the greater cross-education effect following eccentric training remains unknown and untested.

Given lack of muscle hypertrophy in the untrained limb (Farthing and Chilibeck, 2003), along with reports of increased corticospinal excitability (Kidgell et al., 2011; Goodwill et al., 2012), reduced corticospinal inhibition (Latella et al., 2012), reduced interhemispheric inhibition (IHI) (Hortobágyi et al., 2011) and increased voluntary activation (Lee et al., 2009), cross-education is believed to occur as a result of neural adaptations. While direct evidence to substantiate such claims is increasing, the exact mechanisms and specific locus of adaptation for the

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Abbreviations: ANOVA, analysis of variance; FCR, flexor carpi radialis; GABA_A, γ -aminobutyric acid; ICI, intracortical inhibition; IHI, interhemispheric inhibition; iM1, ipsilateral primary motor cortex; ISI, inter-stimulus interval; MEP amplitude, motor-evoked potential; M_{MAX} , maximum compound wave; MVIC, maximal voluntary isometric contraction force; rmsEMG, root mean square electromyography; RMT/AMT, resting and active motor thresholds; sEMG, surface electromyography; SICl, short-interval intracortical inhibition; SP, silent period; TMS, transcranial magnetic stimulation..

cross-education of strength remain unresolved (Carroll et al., 2006; Ruddy and Carson, 2013).

Two theories for the potential mechanism underpinning cross-education have been presented (Ruddy and Carson, 2013 for detailed review). Firstly, the ‘bilateral-access’ hypothesis involves the development of motor engrams (i.e. reorganization of movement representations within the M1) following unilateral practice, that can be accessed not only by the trained limb, but also by the untrained limb. The second ‘cross-activation’ hypothesis is based on the concept of unilateral contractions being driven by bilateral cortical activity in both the contralateral M1 and the ipsilateral primary motor cortex (iM1), producing lasting neuroplastic changes in both cortices. Certainly, transcranial magnetic stimulation (TMS) studies have shown that bilateral corticospinal excitability is facilitated by high-force contractions, with the scale of ipsilateral corticospinal effects being relative to the level of force gradation (Dettmers et al., 1995; Muellbacher et al., 2000; Hortobágyi et al., 2003). However, the neural adaptation underpinning cross-education following unilateral eccentric training remains unknown.

The corticospinal control of eccentric contractions are organized differently to concentric contractions (Enoka, 1996; Sekiguchi et al., 2003; Gruber et al., 2009; Howatson et al., 2011), with the duration of the silent period (SP) being reduced during eccentric contractions compared to concentric (Sekiguchi et al., 2003), intracortical inhibition (ICI) is significantly reduced, while intracortical facilitation (ICF) is increased during forceful eccentric contractions, but not during concentric contractions (Howatson et al., 2011). Cortical excitability of the iM1 is facilitated during eccentric contractions of the right wrist flexors compared to concentric contractions (Howatson et al., 2011). Taken together, compared to concentric contractions, cortical excitability is facilitated in both contralateral and iM1’s during eccentric contractions and the neural networks involved in ICI and IHI appear to be influenced by the type of contraction. On this basis ICI and IHI might be the primary mechanism underpinning the cross-education effects following eccentric training compared to concentric, however, this remains to be tested.

Cross-education has gained scientific and clinical interest, primarily for the potential to minimise strength loss and enhance recovery in patients that are unable to perform training due to single limb injury or impairment (Farthing et al., 2009). Unilateral training of the free limb

has been found to maintain strength, attenuate muscle atrophy and function of the untrained limb following periods of immobilization (Farthing et al., 2009; Magnus et al., 2010) and fracture (Magnus et al., 2013). Given the clinical relevance of cross-education, the purpose of the present study was to determine whether the TMS responses following eccentric or concentric cross-education training are different and whether this difference may explain the change in strength of the untrained limb. We hypothesized that unilateral eccentric training would provide a greater stimulus to the non-exercising limb than concentric training and these behavioral changes in strength would be accompanied with modulation in neurophysiological indices associated with cross-education.

EXPERIMENTAL PROCEDURES

Participants

Twenty-seven participants (15 males aged 25 ± 1 years and 12 females aged 27 ± 2 years) were selected on a voluntary basis. All volunteers provided written informed consent prior to participation in the study, which was approved by the Human Research Ethics Committee in accordance to the standards established by the Declaration of Helsinki. All participants were right-hand dominant as determined by the Edinburgh Handedness Inventory (Oldfield, 1971), had not participated in strength training for at least 12 months, and were free from any known history of peripheral or neurological impairment. Prior to the experiment, all participants completed the adult safety screening questionnaire to determine their suitability for TMS (Keel et al., 2001).

Experimental approach

Fig. 1 outlines the organization of the study. Participants were required to attend a familiarization session that involved performing five eccentric, concentric and isometric contractions of the right and left wrist flexors along with exposure to single-pulse TMS. Following the familiarization session, participants were randomly (based strength) allocated to a control, eccentric training or concentric training group. All participants underwent TMS, ultrasonography, and maximum strength testing (isometric, eccentric and concentric) before and after a 4-week supervised strength-training program; however control participants only undertook pre- and post-testing.

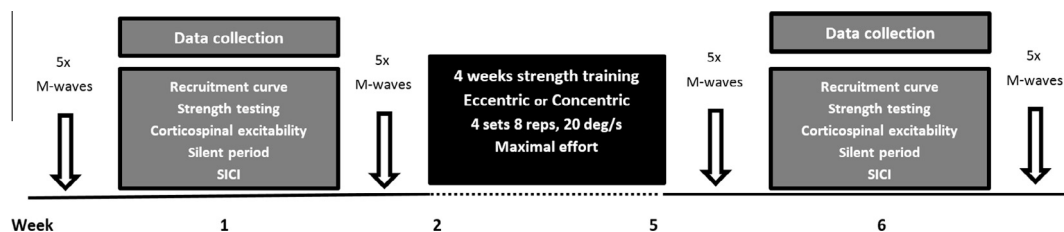


Fig. 1. Schematic representation of the experimental design with measures obtained pre and following 4 weeks of maximal unilateral eccentric or concentric strength training of right wrist flexors. Pre and post measures included assessment of peripheral muscle excitability (M-waves), corticospinal excitability and inhibition recruitment curves, short interval intracortical inhibition (SICI) and muscle strength.

Post-testing was carried out between 36 and 48 h after the final training session.

Muscle thickness

Thickness of the right trained and left untrained wrist flexors (a combined measure of the anterior forearm musculature in cm) was measured with a portable ultrasound device (Sonosite Ultrasound, Springfield, NJ, USA) after a protocol adapted from Magnus et al., (2010). The site of measurement was obtained while the participant rested their forearm on a bench in a supinated position, with the elbow flexed at 90°. A 8- to 15-Hz transducer probe was lubricated with transmission gel and placed lightly on the marked area of the skin, which was 5 cm distal to the left and right olecranon ensuring minimal compression of the muscle before measurement. This placement of the transducer provides a reliable cross-sectional view of the anterior forearm musculature (Noto et al., 2014). The average of six readings served as the final value for muscle thickness.

Voluntary strength testing

Maximal voluntary isometric contraction force (MVIC) of the trained right and untrained left wrist flexors was determined on an isokinetic dynamometer (Biodex System 4 Pro, Biodex Medical Systems, Shirley, USA). Participants were seated in the isokinetic dynamometer, shoulders relaxed and elbow flexed at 90°, with the forearm supinated and fastened firmly on the arm rest. The dynamometer attachment was positioned such that the radial styloid process was aligned with the axis of rotation of the dynamometer. The researcher instructed them to flex the wrist as forcefully as possible by pressing the palm of the hand into the dynamometer attachment. Three trials were performed, each trial was 3 s in duration and separated by a 3-min rest to minimise fatigue. The order of testing the right or left limb was randomized. The greatest recorded force output served as the participants MVIC. For the left untrained limb, the greatest force output was used to determine the target torque levels (5%, 20% and 40% MVIC torque) to be maintained during TMS trials. Participants also performed 3 maximal eccentric and concentric contractions of the right trained and left untrained wrist flexors using the isokinetic dynamometer. Relative to the 0° neutral position, eccentric contractions commenced at 20° of wrist flexion and the participants were instructed to maximally resist the dynamometer by gripping the dynamometer handle. Concentric contractions commenced with the wrist in 20° of extension. Contraction velocity was set at 20°/s and data acquisition was initiated when the participant applied 1% of maximal voluntary force to the dynamometer handle. The largest recorded force output obtained during lengthening and shortening contractions served as the participant's maximum. Again, for the left untrained wrist flexors, the largest recorded force output obtained was used to determine the target torque level of 40% eccentric and concentric torque to be maintained during the TMS trials and visual feedback of the target

torque was displayed on the Biodex computer monitor. The order of strength testing (left/right wrist, isometric/concentric/eccentric) tests were counterbalanced, but consistent within participants.

Surface electromyography (sEMG)

The area of electrode placement was shaven to remove fine hair, rubbed with an abrasive skin rasp to remove dead skin, and then cleaned with 70% isopropyl alcohol. The exact sites were marked with a permanent marker by tracing around the electrode, and this was maintained for the entire 4-week period by both the researcher and participant to ensure consistency of electrode placement relative to the innervation zone. An impedance meter was used to ensure impedance did not exceed 10 k Ω prior to testing. sEMG activity was recorded from the left and right flexor carpi radialis (FCR) muscle using bipolar Ag–AgCl electrodes (8 mm diameter, model E258S; Biopac, Goleta, CA, USA). These electrodes were placed on the FCR muscle, with an inter-electrode distance (center to center) of 2 cm. Specifically, the electrodes were positioned 8–9 cm from the medial epicondyle. A grounding strap placed around the wrist was used as a common reference for all electrodes. sEMG signals were amplified (1000 $\times$), bandpass filtered (high pass at 13 Hz, low pass at 1000 Hz), digitized online at 2 kHz, recorded and analyzed using PowerLab 4/35 (ADInstruments, Bella Vista, Australia).

TMS

TMS was delivered using two Magstim 200² stimulators (Magstim Co, Dyfed, UK) connected via a Bistim unit and a single figure of eight coil (external diameter of each loop 70 mm). The motor hotspot for only the left untrained FCR (with posterior-to anterior-induced current flow in the cortex) was determined and resting and active motor thresholds (RMT/AMT) were established as the intensity at which at least five of 10 stimuli produced motor-evoked potential (MEP) amplitudes of greater than 50 μ V for RMT and greater than 200 μ V for AMT in the left untrained FCR muscle (Sale and Semmler, 2005). To ensure all stimuli were delivered to the optimal motor hotspot throughout testing, participants wore a tight-fitting cap marked with a latitude-longitude matrix, positioned with reference to the nasion-inion and interaural lines.

Single-pulse recruitment curves were collected during low-level isometric contractions of the left untrained wrist flexors. Low-level contractions equated to $5 \pm 2\%$ of root mean square electromyography (rmsEMG) during MVIC (see Table 1) and were performed by maintaining the wrist and fingers in a straight position. This level of background electromyography (EMG) has been previously used to produce reliable MEPs and SP durations (Sale and Semmler, 2005, See Fig. 2). Consistent muscle activation was confirmed by recording pre-stimulus rmsEMG throughout the session. For a single recruitment curve, five stimuli were delivered at 90%, 110%, 130%, 150%, 170%, 190%, 210% and

Table 1. Mean ($\pm$ SE) for SICI at 40% MVIC, SP and PP TMS pre-stimulus rmsEMG at 120% AMT and muscle thickness for the untrained limb

Group	SICI 40% MVIC		SP rmsEMG (%MVIC _{MAX})		PP rmsEMG (%MVIC _{MAX})		Muscle thickness (cm)	
	Pre	Post	Pre	Post	Post	Post	Pre	Post
Control	77.0 $\pm$ 5.6	77.3 $\pm$ 5.4	1.9 $\pm$ 0.3	2.4 $\pm$ 0.2	2.4 $\pm$ 0.2	2.5 $\pm$ 0.3	2.7 $\pm$ 0.3	3.2 $\pm$ 0.4
Eccentric	66.3 $\pm$ 5.9	84.8 $\pm$ 4.3 ^{*†#}	3.3 $\pm$ 0.6	2.5 $\pm$ 0.2	2.5 $\pm$ 0.2	3.6 $\pm$ 0.9	3.9 $\pm$ 0.6	4.1 $\pm$ 0.8
Concentric	73.6 $\pm$ 4.0	74.1 $\pm$ 5.1	4.0 $\pm$ 0.7	2.3 $\pm$ 0.5	2.4 $\pm$ 0.5	3.9 $\pm$ 0.6	4.1 $\pm$ 1.3	3.4 $\pm$ 0.8

MVIC, maximal voluntary isometric contraction; SICI, short-interval intracortical inhibition; SP, single-pulse; PP, paired-pulse; TMS, transcranial magnetic stimulation.

* Denotes significant to control group ($P > 0.05$).

† Denotes significant to concentric group ($P > 0.05$).

Denotes significant to pre ($P > 0.05$).

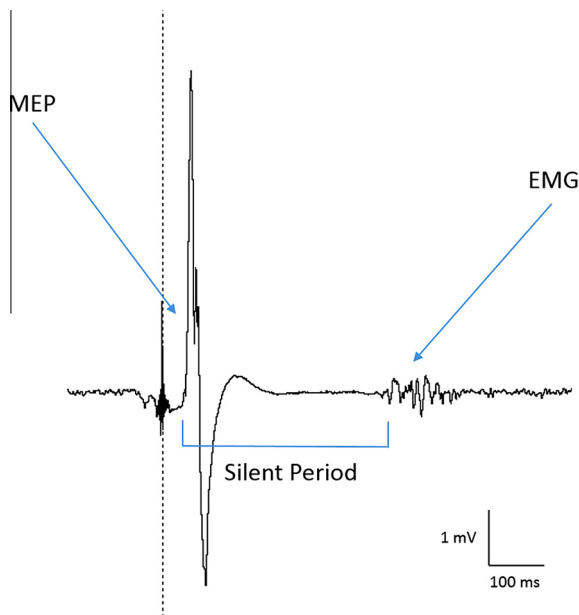


Fig. 2. Sample recording of a silent period from the left untrained forearm flexors during a low-level contraction ($5 \pm 2\%$ maximal rmsEMG) at baseline. Silent period was measured from the onset of the MEP (cursor 1) to the return of EMG (cursor 2).

230% of AMT. In addition, corticospinal excitability and inhibition (SP duration) was quantified, with a test intensity of 120% AMT (adjusted only if there was a change in AMT following the intervention), in five pre-determined target force levels (5% MVIC, 20% MVIC, 40% MVIC, 40% eccentric torque and 40% concentric torque for only the left untrained wrist flexors). Eight TMS stimuli were applied at each target torque level for only the untrained left wrist flexors, which was kept consistent between pre- and post-testing. During eccentric and concentric contractions, TMS stimuli were automatically presented as the left untrained wrist passed through 0° of wrist flexion or extension to ensure that all TMS data were collected at the same anatomical position (Howatson et al., 2011). To control for background sEMG prior to TMS stimulation, all MEPs obtained during eccentric and concentric contractions post training were obtained at the pre torque level. A purpose made Excel macro was used to randomize the TMS trials in blocks of eight contractions across the five-target torque levels (Rantalainen et al., 2013).

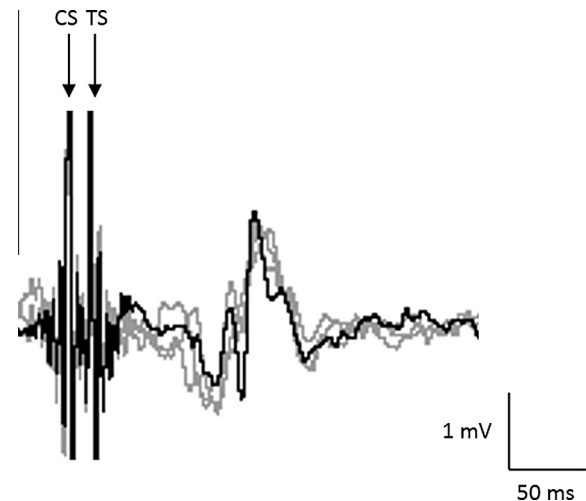


Fig. 3. Five overlaid sweeps of paired-pulse MEPs from a single participant at baseline during a low-level contraction ($5 \pm 2\%$ maximal rmsEMG) of the left untrained wrist flexors. The arrows indicate the triggering of paired-pulse stimulation with the first trigger being the conditioning subthreshold (CS) stimulation, with the second trigger being the suprathreshold (TS) stimulus in the paired-pulse stimulations.

To quantify short-interval intracortical inhibition (SICI), five single-pulse stimuli and five short-interval paired-pulse stimuli were delivered in a random order (Rantalainen et al., 2013; see Fig. 3). The stimulator output intensity was set at 120% of AMT, which was determined during familiarization and adjusted if there was a change in AMT following training. The conditioning stimulus for paired-pulse stimulation was set at 80% of AMT, the inter-stimulus interval (ISI) was 3 ms, and subsequent posterior to anterior current flow was used (Rantalainen et al., 2013). SICI was only obtained for the untrained left wrist flexors.

Maximal compound muscle action potential

Direct muscle responses were obtained from the left and right FCR muscle by supramaximal electrical stimulation (pulse width 200 μ s) of the median nerve under resting conditions (DS7A, Digitimer, Hertfordshire, UK). The site of stimulation that produced the largest M-wave was located by positioning the bipolar electrodes in the cubital fossa. An increase in current strength was applied to the median nerve until there was no further

increase observed in the amplitude of the sEMG response (M_{MAX}). To ensure maximal responses, the current was increased an additional 20% and the average M_{MAX} was obtained from five stimuli, with a period of 6–9 s separating each stimulus. M_{MAX} was recorded at baseline and following the training intervention, to control for possible changes in peripheral muscle excitability that could influence MEP amplitude.

Strength-training protocol

Participants allocated to the eccentric and concentric training groups participated in strength training three times per week of the right wrist flexors (12 sessions in total) on non-consecutive days, for 4-weeks which required either maximal eccentric or concentric contractions performed at 20°/s on an isokinetic dynamometer. Training was performed in a seated position with the forearm supinated and fastened firmly on the arm rest, with posture identical to pre and post strength-testing procedures. The training took place under supervision, and consisted of four sets of 6–8 maximal concentric or eccentric repetitions, with a 3-min recovery between sets. The control group continued performing typical daily activities without undertaking any additional training. During all supervised training sessions, EMG activity was recorded from the left untrained wrist flexors to determine the magnitude of mirror EMG activity as a percent of maximum rmsEMG obtained during maximum eccentric and concentric testing of the left untrained limb that was obtained at baseline testing.

Data analyses

Pre-stimulus rmsEMG. Pre-stimulus rmsEMG activity was determined in the untrained left wrist flexors 100 ms prior to each TMS stimulus during pre and post testing. Any trial in which pre-stimulus rmsEMG exceeded $5 \pm 2\%$ of maximal rmsEMG were discarded and the trial repeated.

Cortical excitability. The peak-to-peak amplitude of MEPs evoked as a result of stimulation was measured only in the left untrained FCR muscle contralateral to the cortex being stimulated in the period 10–50 ms after stimulation. MEP amplitudes were analyzed (LabChart 8 software, ADInstruments, Bella Vista, NSW, Australia) after each stimulus was automatically flagged with a cursor, providing peak-to-peak values in μV , averaged and normalized to the M_{MAX} and multiplied by 100. To construct single-pulse recruitment curves, stimulus intensity was plotted against MEP amplitude, and then fitted with a non-linear Boltzmann equation according to previously defined protocols (Devanne et al., 1997).

SICI. SICI was measured (only for the untrained left wrist flexors) with a conditioning-test pulse procedure with an ISI of 3 ms. The test stimulus was set at 120% of AMT, whereas the conditioning stimulus was set at

80% of AMT. The conditioned MEP amplitude was expressed as a percentage of the unconditioned test MEP amplitude to calculate the level of ICI.

Corticospinal inhibition. Following the determination of maximum force output for only the left untrained wrist flexors, participants were asked to produce an isometric contraction at 5%, 20% and 40% of their the left wrist flexor MVIC and at 40% of their eccentric and concentric torque. While contracting the left untrained wrist flexors, a TMS pulse was delivered over the right motor cortex, at 120% AMT, to evoke a SP in the left FCR. All post measures were obtained at the pre torque level, as increases in background EMG as a result of training could confound MEP amplitudes. The average duration of the SP over the eight trials was calculated. The beginning of the MEP preceding the SP and the subsequent onset of the EMG signal were manually selected and used to indicate the beginning and end of the SP, respectively (see Fig. 2). All processing was completed by the same investigator who was blinded to each condition (Christie and Kamen, 2014).

Sample size calculations and statistical analysis

The number of participants required was based on power calculations for the expected changes in mean rectified MEPs (sEMG recordings from the hand muscles). As this is innovative work without historical precedent, estimates of sample size are difficult. However, using previous cross-education data in healthy untrained adults (Carroll et al., 2006), we estimated that six participants in each group would provide at least 80% power (95% confidence interval) to detect a 15% difference in mean rectified MEPs assuming a SD of 10–15% between groups at $P < 0.05$ (two-tailed).

All data were screened with the Brown–Forsythe test and found to be normally distributed (all $P > 0.05$) and thus the assumptions of the analysis of variance (ANOVA) were not violated. Subsequently, parametric analysis using a mixed factorial ANOVA appropriate for a 3×2 design [three groups (control, eccentric, concentric), two time points (pre-testing, post-testing)] comparing multiple outcome measures (muscle thickness, MVIC torque, eccentric torque, concentric torque, pre-stimulus EMG, mirror activity, corticospinal excitability, SICI, and SP). Univariate and post hoc (LSD) analysis for each dependent measure followed where significant multivariate effects were found. SPSS version 22.0 (SPSS Inc., Chicago, Ill, USA) was used for all statistical analyses, with the level of significance used for all tests set at $P < 0.05$. All data are presented as mean $\pm$ standard error (SE).

RESULTS

Pre-stimulus rmsEMG and maximal compound wave

Pre-stimulus rmsEMG did not vary between single- and paired-pulse trials, and there were no changes over time for any group and thus no group by time interactions were present (Table 1). Similarly, M_{MAX} did not change

as a function of time or by group, and there were no group by time interactions present (all $P > 0.05$). Mean M_{MAX} for the left untrained wrist flexors for control, eccentric and concentric pre training were 7.802 ± 1.303 mV, 7.837 ± 0.948 mV and 9.251 ± 1.222 mV, respectively. Mean M_{MAX} post training were 7.632 ± 1.238 mV, 7.458 ± 1.026 mV and 8.506 ± 1.237 mV, respectively for control, eccentric and concentric groups.

Isometric, eccentric and concentric torque of the untrained limb

Fig. 4A–C shows the change in contralateral forearm muscle strength. There were no differences in baseline isometric, eccentric or concentric strength of the untrained left wrist flexors between groups (all $P > 0.05$). Following the intervention there was a main

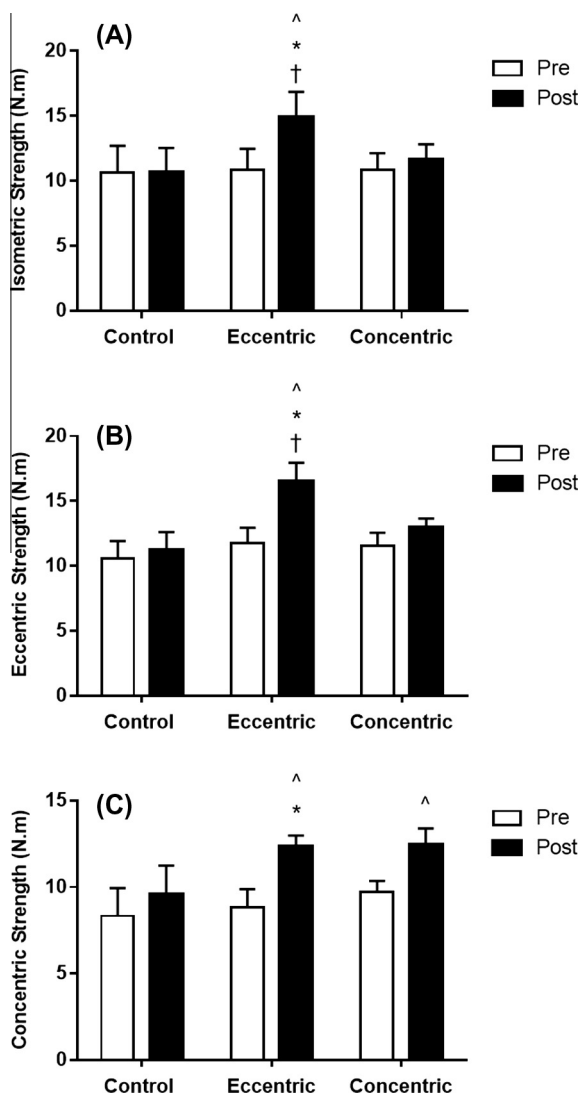


Fig. 4. (A)–(C) Mean ($\pm$ SE) changes in isometric (A), eccentric (B) and concentric (C) strength of the left untrained forearm flexors following 4-weeks of maximal eccentric or concentric strength training of the right forearm flexors or control (no training). ^{*}Denotes significant to control group ($P < 0.05$); [†]denotes significant to concentric group ($P < 0.05$); [#]denotes significant to pre ($P < 0.05$).

effect for time ($P < 0.001$) and a group by time interaction for isometric strength ($P < 0.001$). For the untrained limb, eccentric training resulted in a 43% increase in isometric strength, which was significantly different to the concentric training group (11%, $P = 0.003$) and the control group (2%, $P < 0.001$). Furthermore, there was a main effect for time ($P < 0.001$) and a group by time interaction ($P = 0.002$) for eccentric strength of the left untrained wrist flexors. The eccentric training group, increased eccentric strength of the untrained wrist flexors by 47% compared to the concentric training group (14%, $P = 0.014$) and the control group (7%, $P = 0.003$). Following the intervention there was also a main effect for time ($P < 0.001$) and a group by time interaction ($P = 0.040$) for concentric strength of the untrained wrist flexors. The concentric training group increased their concentric strength of the untrained wrist flexors by 28%, which was significantly less compared to the eccentric training group, which increased their concentric strength by 49% ($P = 0.034$). There were no non-significant difference in concentric strength between the concentric training group and the control group (18%, $P > 0.05$).

Muscle thickness

There were no differences in thickness of the untrained wrist flexors between groups at baseline ($P > 0.05$). There were no main effects for time ($P = 0.063$) or group by time interactions ($P = 0.725$) detected following the training intervention (Table 1). There were also no differences in muscle thickness of the right trained wrist flexors between groups at baseline ($P > 0.05$), or no main effects for time ($P = 0.095$) or group by time interactions ($P = 0.935$) detected for the trained right wrist flexors following the intervention.

Corticospinal excitability

MEP amplitudes (expressed as a percentage of M_{MAX}) were obtained at different levels of MVIC (5%, 20% and 40%) and during 40% of eccentric and concentric torque. MEP amplitudes were similar between control, eccentric training and concentric training groups at

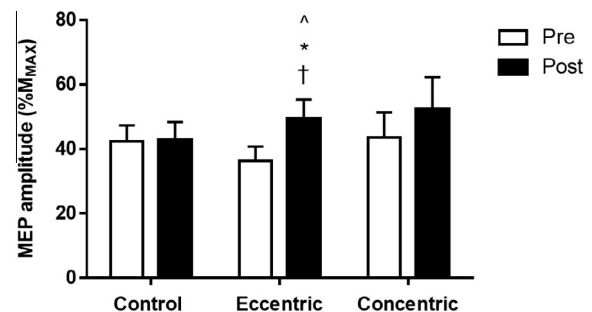


Fig. 5. Mean ($\pm$ SE) change in MEP amplitude (for the left untrained wrist flexor at 40% above AMT) following 4-weeks of maximal eccentric or concentric strength training of the right wrist flexors. ^{*}Denotes significant to control group ($P < 0.05$); [†]denotes significant to concentric group ($P < 0.05$); [#]denotes significant to pre ($P < 0.05$).

baseline ($P > 0.05$). We observed no main effects for time or group by time interactions for any change in MEP amplitudes between 5%, 20% and 40% of isometric torque (all $P > 0.05$) for the untrained limb. However, at 40% of eccentric torque we observed a significant main effect for time ($P = 0.003$) and a group by time interaction ($P = 0.012$) (Fig. 5). Eccentric strength training of the right limb, resulted in a 51% increase in MEP amplitude of the left untrained limb (expressed as percentage of M_{MAX}) during eccentric contractions compared to a 13% increase in the concentric training group ($P = 0.033$) and the 10% increase in the control group ($P = 0.005$). There were no changes in MEP amplitudes for the left untrained limb at 40% concentric torque for any groups (all $P > 0.05$).

Corticospinal inhibition

Corticospinal inhibition was assessed with the duration of the SP, which was obtained at different levels of isometric torque (5%, 20%, and 40%) and during 40% of eccentric and concentric torque. The SP durations were similar between control, eccentric training and concentric training at baseline (all $P > 0.05$). Following the intervention, at 5% of isometric torque, there was a main effect for time ($P = 0.032$) and a group by time interaction ($P = 0.004$). For the untrained left wrist flexors, eccentric strength training reduced the SP duration by 27% compared to 4% for the control group ($P = 0.001$) and 4% for the concentric training group ($P = 0.006$). At 20% of isometric torque for the left untrained wrist flexors, there was a main effect for time ($P = 0.043$), with eccentric training reducing the SP by 15% however; this reduction was not different to concentric training (8%) or the control group (1%, $P > 0.05$).

Interestingly, we observed several main effects for time and group by time interactions for a reduction in SP duration at a number of stimulus intensities above AMT for the untrained left wrist flexors (Fig. 6A–C). At 110% of AMT, we observed a main effect for time ($P = 0.002$) and a group by time interaction ($P = 0.006$). Following the intervention, SP duration reduced by 27% following eccentric training compared to 1% in the concentric training group ($P = 0.004$) and compared to 2% in the control group ($P = 0.008$). A similar effect was observed at 130% AMT, with eccentric training reducing the duration of the SP by 21% compared to 5% in the concentric training group ($P = 0.048$) and compared to 5% for the control group ($P = 0.005$). At 150% AMT, eccentric training reduced the SP duration by 15% compared to the control group (1%, $P = 0.021$), but this change was not different to the concentric training group (1%). At 170% AMT, only eccentric training reduced (10%) the duration of the SP ($P = 0.006$), however this reduction was not different to the concentric training group (3%) or control group (1%). There were no differences in the input and output properties of the corticospinal tract following the training intervention for any group (all $P > 0.05$).

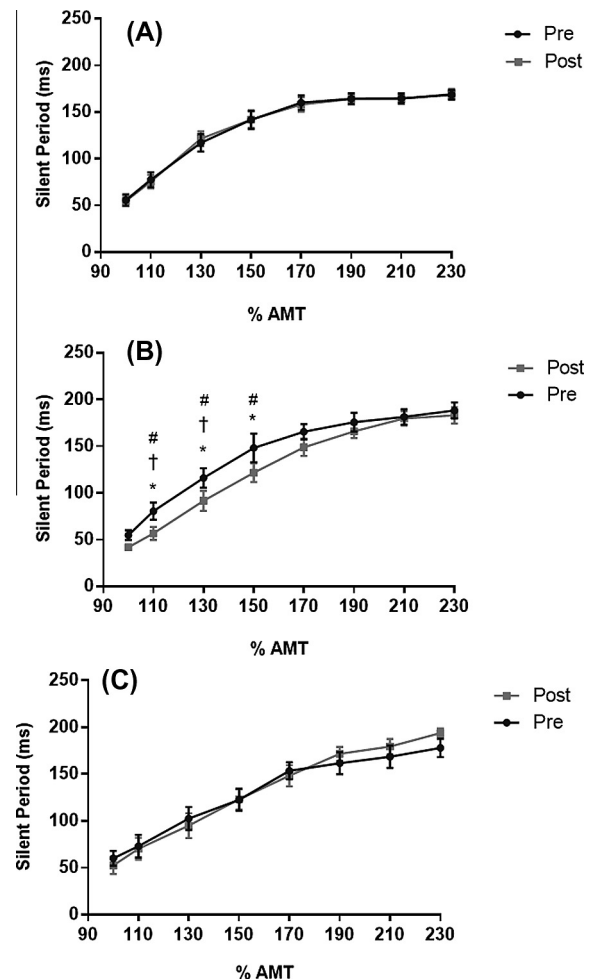


Fig. 6. (A)–(C) Mean ($\pm$ SE) changes in cortical silent period duration (CSP) of the ipsilateral motor cortex (M1) following 4-weeks of control (A – no training) eccentric (B) or concentric (C) strength training of the right wrist flexors. *Denotes significant to control group ($P < 0.05$); † denotes significant to concentric group ($P < 0.05$); #denotes significant to pre ($P < 0.05$).

SICI

SICI measures were obtained at a range of isometric torque levels (5%, 20%, 40% MVIC), and during 40% of eccentric and concentric torque prior to and following the 4-week training intervention. At 5% and 20% of MVIC there were no main effects for the release of SICI in any group (all $P > 0.05$). However, for the untrained left wrist flexors, there was a group-by-time interaction ($P = 0.009$) for a release of SICI during 40% of isometric torque (Table 1). At 40% of MVIC, a group-by-time interaction was observed ($P = 0.007$), with eccentric training reducing SICI by 32%, which was different to the concentric training group (2%, $P = 0.003$) and the control group (1%, $P = 0.002$). No changes in SICI were observed at 40% eccentric and concentric torque for any groups (all $P > 0.05$).

Mirror EMG activity

We also examined whether maximal concentric or eccentric contractions during training affected the

degree of mirror activity in the contralateral untrained left wrist flexors. Averaged across all training weeks there were no differences in mirror EMG activity for the eccentric training group ($1.33 \pm 0.22\%$ rmsEMG_{max}) or concentric training group ($1.5 \pm 0.37\%$ rmsEMG_{max}, $P > 0.05$).

DISCUSSION

This is the first study to examine the TMS responses of the untrained contralateral limb, following 4-weeks of either eccentric or concentric cross-education strength training. The main findings of the current experiment were as follows: (1) Eccentric strength training of the right wrist flexors increased strength of the left untrained wrist flexors more than concentric strength training. Eccentric strength training increased both isometric and dynamic strength of the untrained wrist flexors, (2) eccentric strength training reduced cortical inhibition of the untrained wrist flexors, as determined by SICI and the duration of the SP and (3) corticospinal excitability of the untrained wrist flexors was only increased during 40% of torque during eccentric contractions following eccentric training compared to concentric training.

Eccentric strength training enhances the cross-transfer of strength

We examined the hypothesis that eccentric compared to concentric strength training would produce greater changes in muscle strength of the untrained limb because of its unique activation and greater training effects on the neuromuscular system (Enoka, 1996; Hortobágyi et al., 1996, 1997; Howatson et al., 2011). Several short-term strength-training studies have demonstrated greater changes in muscle strength following eccentric training when compared to concentric training (Hortobágyi et al., 1996; Komi and Buskirk, 1972; Pavone and Moffat, 1985). Consequently, the cross-transfer of strength following eccentric training was greater following eccentric training which supports the recent finding by Lepley and Palmeri-Smith (2014).

The cross-education effect in the current study was prominent and markedly different between groups, with eccentric training resulting in a 47% transfer of eccentric strength, 43% for isometric and 49% for concentric, with concentric training resulting in significantly less cross-transfer effects (eccentric 14%, isometric 11% and concentric 28%). The large increase in strength of the untrained limb following eccentric training is comparable to an earlier study that reported a 77% increase in eccentric strength compared to only 30% following concentric training (Hortobágyi et al., 1997). However, it should be noted, that in the present study, both the concentric and eccentric training groups performed maximal voluntary contractions of an upper limb muscle throughout the training period, whereas previous studies have used submaximal contractions of lower limb muscles (Johnson et al., 1976; Dudley et al., 1991). Although, during maximal unilateral muscle contractions, the untrained muscle can exhibit up to greater than 20% of mirror EMG activity (Zijdewind et al., 2006), in the present study,

mirror EMG activity in both the concentric and eccentric training conditions, was only 1.5% of maximum EMG activity. Given the low level of mirror EMG activity and no detectable changes in muscle thickness, the change in muscle strength appears to be due to facilitation of the corticospinal pathway innervating the untrained limb, which was different between eccentric and concentric training.

Eccentric training differentially modulates corticospinal inhibition and excitability

Several lines of evidence support the view that GABA_A-mediated ICI contributes to M1 plasticity (Werhahn et al., 1999). The present results demonstrate a task-specific effect for a reduction of SICI confined to the corticospinal tract innervating the untrained limb (i.e. iM1) following eccentric training only. This finding is consistent with previous experiments showing that SICI is acutely reduced in the iM1 during eccentric compared to concentric contractions (Howatson et al., 2011). In the current study, eccentric training of the right limb reduced, evidenced by reduced SICI, the synaptic efficacy of GABA_A receptors of neurons forming cortico-cortical networks within the untrained M1, releasing pyramidal neurons from inhibition (Kujirai et al., 1993). Therefore, there appears to be a task-dependant modulation of SICI, which supports previous observations (Perez and Cohen, 2008; Howatson et al., 2011) demonstrating eccentric resistance exercise uniquely modulating SICI following unilateral eccentric training. Further, given that mirror EMG throughout training was not different between concentric and eccentric training, the reduction in SICI, confined to the iM1, appears to be due to task-dependant effects of chronic eccentric contractions. In regard to the cross-education phenomena, the current results showed that eccentric training uniquely modulated and reduced ipsilateral SICI and further highlight the effect that different training tasks have on the cross-transfer of strength.

The present results showed that the duration of the SP for the untrained wrist flexors was only reduced following eccentric strength training. Again, there seems to be a task-dependant effect of eccentric training differentially modulating indices of cortical inhibition. The SP that follows the excitatory MEP is caused by activation of long-lasting GABA_B-mediated inhibition and reflects a temporary suppression in motor cortical output (Lang et al., 2006). The reduction in the SP of the untrained limb suggests a task-specific effect that reduced the inhibitory input to the motoneurone pool, which could contribute to the net excitability of the corticospinal tract. The first 50 ms is widely assumed to be mediated by spinal mechanisms associated with after-hyperpolarization of the motoneurons and recurrent inhibition, however after 100-ms reductions in the SP is due to supraspinal inhibition (Inghilleri et al., 1993). The current findings show that the duration of the SP following eccentric training was around 59 ms, which suggests this reduction could be influenced by spinal factors (possibly reflex arcs) evoked by eccentric training (Butler et al., 2011). Importantly, the duration of the SP was only reduced following eccentric

training, which is also consistent with the reduction in SICl following eccentric training. These data show that eccentric training modulated two different neural networks that are involved in inhibition, which collectively act to improve the net excitatory drive to the muscle.

Following the training intervention, MEP amplitude of the untrained wrist flexors increased by 51% only for the eccentric training group during 40% of eccentric torque, while concentric training had no net effect on corticospinal excitability. This finding is consistent with previous work that also demonstrated an acute increase in ipsilateral MEPs during eccentric contractions compared to concentric contractions (Howatson et al., 2011). It seems that unilateral eccentric strength training specifically modulated the ipsilateral corticospinal tract via a reduction in ICI and possibly the aforementioned spinal inhibition, resulting in a net increase in corticospinal excitability. This interpretation is supported by previous cross-education studies (Hortobágyi et al., 2011; Latella et al., 2012) however; the current findings showed that only eccentric training modulated these pathways. Therefore, these findings support a role for the corticospinal tract (ipsilateral to the trained limb) contributing to the contralateral adaptations following unilateral eccentric training (Housh et al., 1996; Hortobágyi et al., 1997; Farthing and Chilibeck, 2003).

There are several limitations to the present study. First, due to the number of moderate intensity muscle contractions performed by each participant, we were limited to only examining the iM1 and corticospinal tract and not the contralateral M1. Second, we only collected stimulus–response curves during very low isometric contractions and thus we were unable to demonstrate the task-dependant effects of contraction mode specifically on the gain of the corticospinal tract following unilateral strength training. Finally, the addition of measures at a segmental level, particularly cervicomedullary MEPs and Hoffman reflex (H-reflex), would provide additional information as to the site of adaptation within the corticospinal tract following unilateral strength training. Despite these limitations, the current data provide new knowledge about the task-specific neural adaptations to unilateral strength training, which may underpin the cross-transfer of strength.

CONCLUSION

The results of the present study are consistent with the working hypothesis that eccentric training of the right limb uniquely modulates the cross-transfer of strength and ipsilateral corticospinal excitability and inhibition compared to concentric training. There is a clear pattern that eccentric training at maximal intensities reduces cortical inhibition and the duration of the SP and thereby increasing the cross-transfer of strength. These findings have important clinical implications as previous research showed that during periods of limb immobilization (Farthing et al., 2011) and wrist fractures (Magnus et al., 2013), unilateral strength training can attenuate the loss of muscle function and atrophy. Importantly, the current study shows that high-effort eccentric training,

results in greater levels of strength transfer that is modulated, at least in part by a reduction in corticospinal inhibition.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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